Report 2: Film Characteristics and Box Office Success

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1 Introduction

Studios commit hundreds of millions of dollars to films before a single ticket is sold, making reliable predictors of success highly valuable. This report asks: **what film characteristics are associated with critical and commercial success among the top 500 highest-grossing domestic films of all time?** We measure critical success using IMDb user rating and commercial success using log-transformed domestic gross, and examine genre, runtime, MPAA rating, release timing, and decade as predictors. As a secondary descriptive question, we also explore whether the 74 book adaptations in our sample (14.8%) differ from other films in ratings or earnings, and whether source material quality-measured via Goodreads ratings-correlates with film reception. The dataset was enriched with metadata from IMDb, TMDb, Wikidata, and the UCSD Goodreads dataset. A key limitation is that all films in this sample are commercially successful by definition (each earned at least $140M domestically), so our analysis captures variation in *degree* of success rather than success versus failure.

2 Data

The base dataset is a ranked list of the 500 highest-grossing domestic films of all time, spanning releases from 1937 to 2026, with the bulk of films concentrated in the 2000s and 2010s. Starting from this ranked list, film-level metadata was collected from four external sources via automated pipelines and joined by title and release year.

IMDb ratings and runtimes were obtained from IMDb's non-commercial bulk data files. Content ratings and release dates were retrieved from the TMDb REST API using a bearer token. Distributor information was pulled from Wikidata via SPARQL queries. Book adaptation status (`is_book`) was determined by cross-referencing each film title against a Wikipedia-sourced list of fiction-to-film adaptations, supplemented by manual verification for borderline cases.

3 Exploratory Data Analysis

3.1 Distributions

IMDb ratings range from 4.2 to 9.1 with mean 7.07. The distribution is fairly concentrated, which is expected since all films in the dataset are commercially successful.

Log domestic gross is approximately normal after transformation.

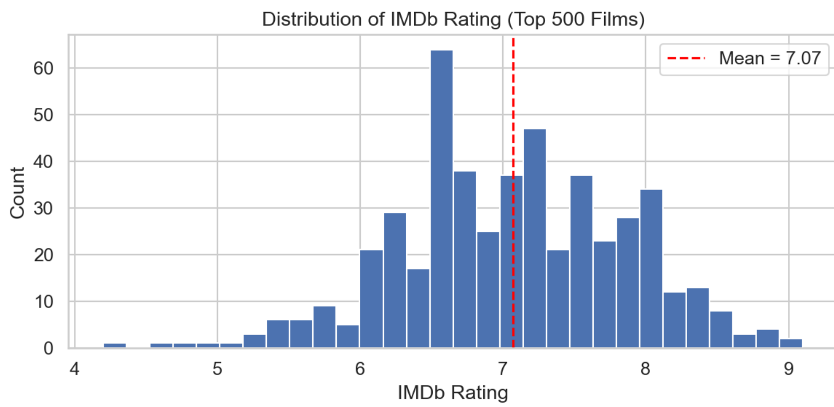

Figure 1: IMDb Rating Distribution

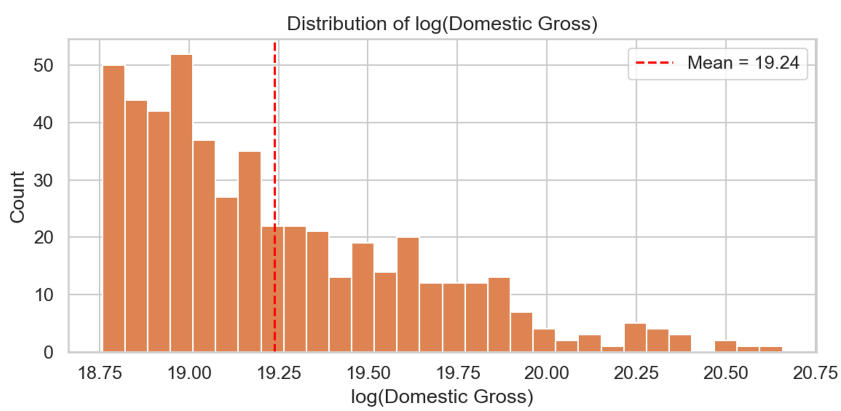

Figure 2: Log Domestic Gross

3.2 Genre

Action films make up over half of the dataset. Drama films have slightly higher average IMDb ratings, while Action and Adventure films dominate box office performance. This suggests that genres associated with large-scale productions tend to generate more revenue.

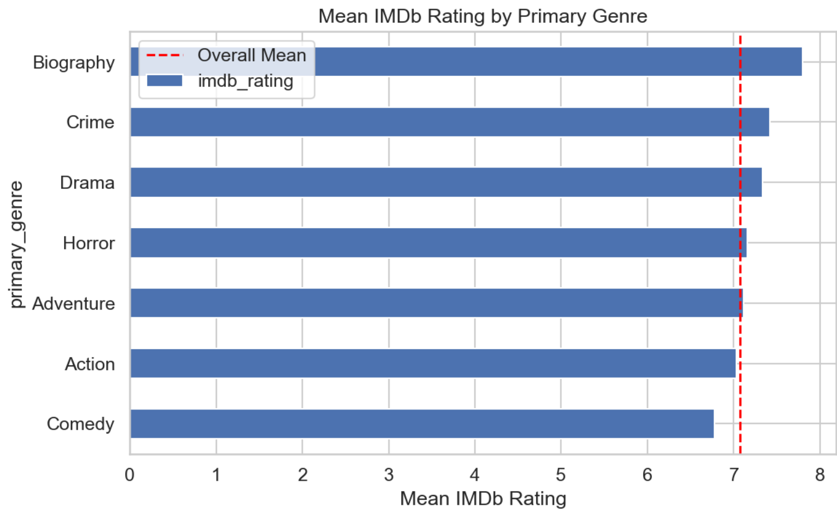

Figure 3: IMDb by Genre

3.3 Runtime

Runtime is the strongest continuous predictor of both outcomes.

- Corr(IMDb, runtime) = 0.27
- Corr(log gross, runtime) = 0.25

Longer films tend to receive higher ratings and earn more. This may reflect the types of films that are given longer runtimes, such as action or epic films, rather than a direct causal effect.

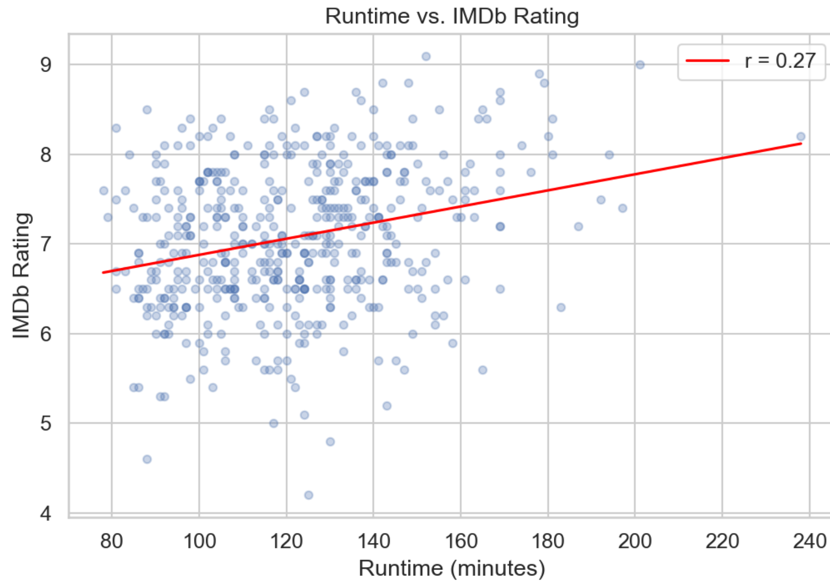

Figure 4: Runtime vs IMDb

3.4 MPAA Rating

G-rated films have the highest average IMDb ratings, although the sample size is small. PG-13 films earn the most at the box office, likely due to broader audience reach. R-rated films earn less on average.

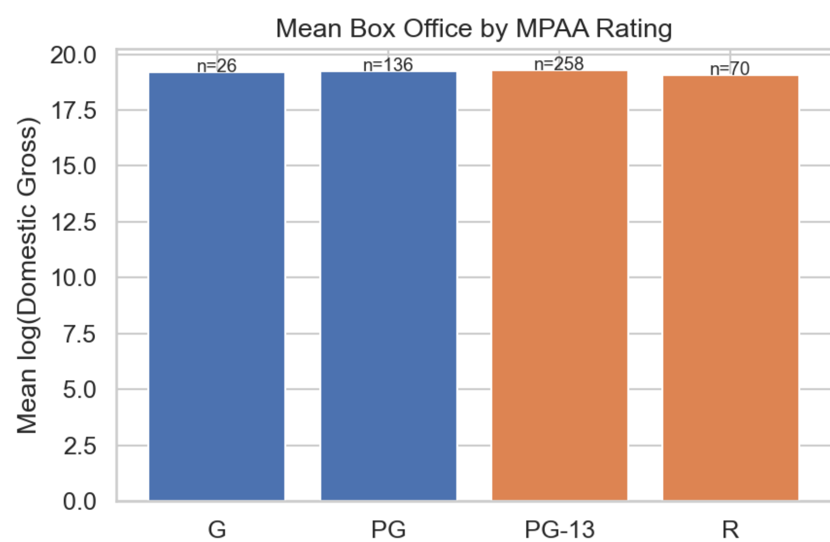

Figure 5: Box Office by MPAA

3.5 Release Month

Summer and holiday months account for most top-grossing films. These periods are traditionally used for major releases. Month has little relationship with IMDb rating and only a modest relationship with revenue.

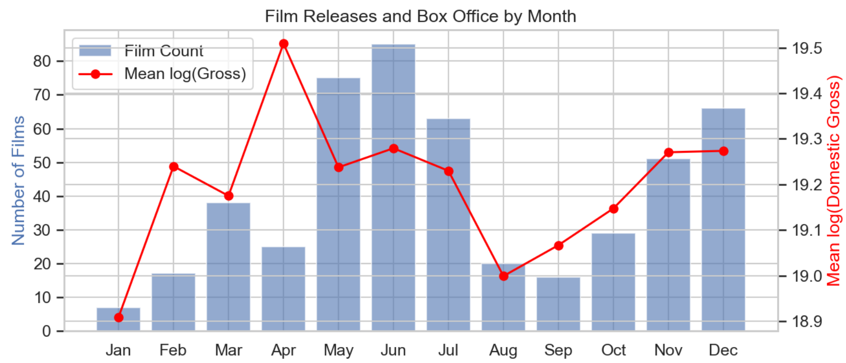

Figure 6: Month and Box Office

3.6 Decade

Older films tend to have slightly higher IMDb ratings, likely due to survivorship bias. More recent films show higher gross in nominal terms.

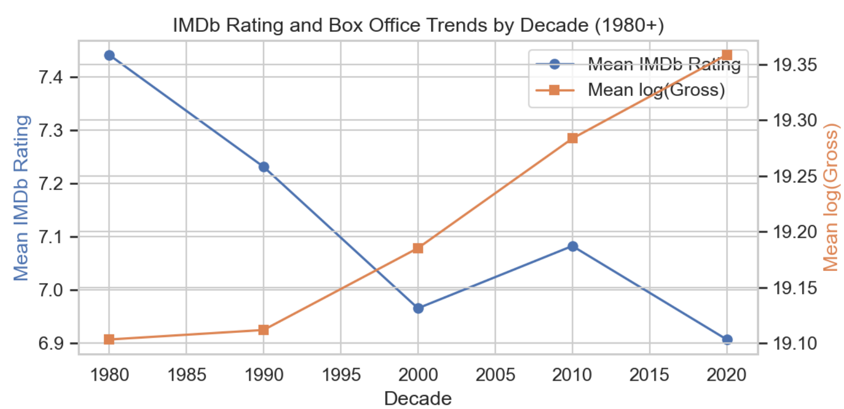

Figure 7: Decade Trends

3.7 Correlation

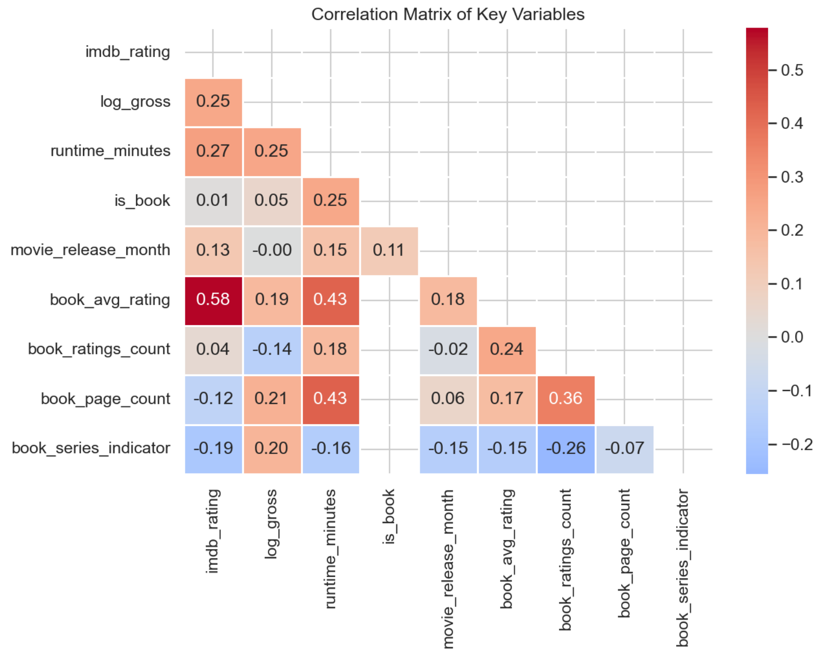

Figure 8: Correlation Heatmap

Runtime, IMDb rating, and log gross show moderate positive relationships. The variable `is_book` has near-zero correlation with both outcomes.

3.8 Book Adaptations

Metric	Book	Other
IMDb Rating	7.09	7.07
Log Gross	19.29	19.23
Runtime	135.7	119.1

Differences between book adaptations and other films are small and not statistically significant. Book adaptations are noticeably longer on average.

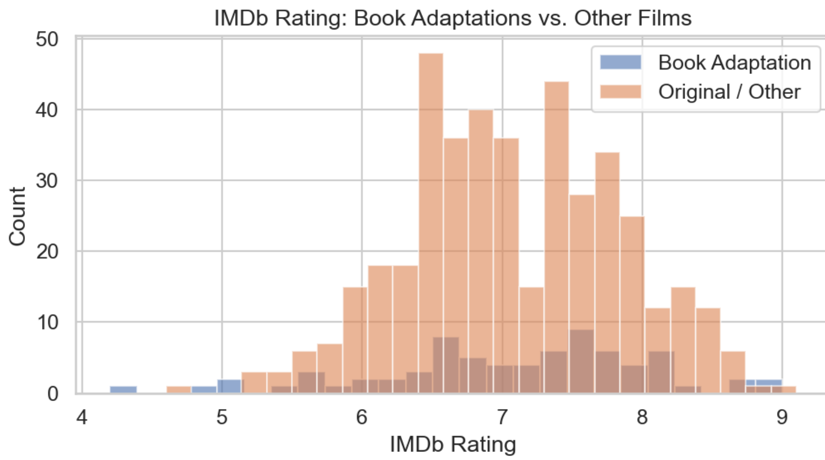

Figure 9: IMDb Comparison

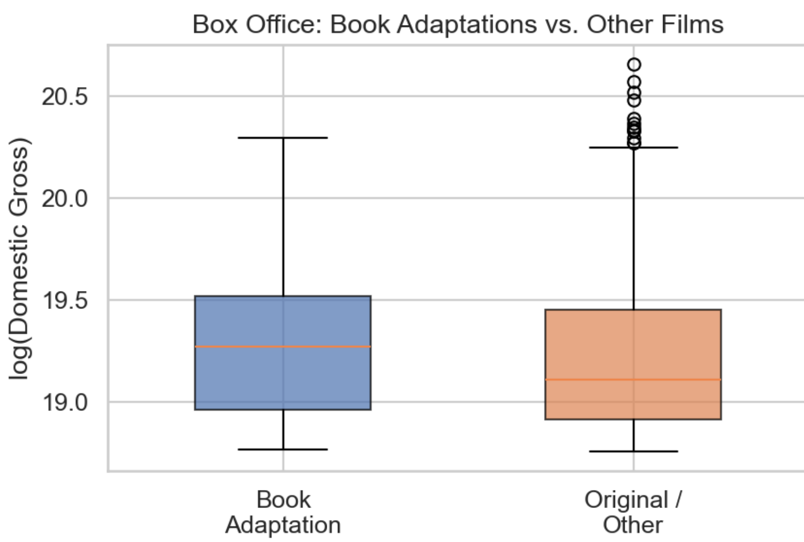

Figure 10: Gross Comparison

Among book adaptations, Goodreads rating correlates with IMDb rating (r = 0.58), suggesting that stronger source material is associated with better audience reception.

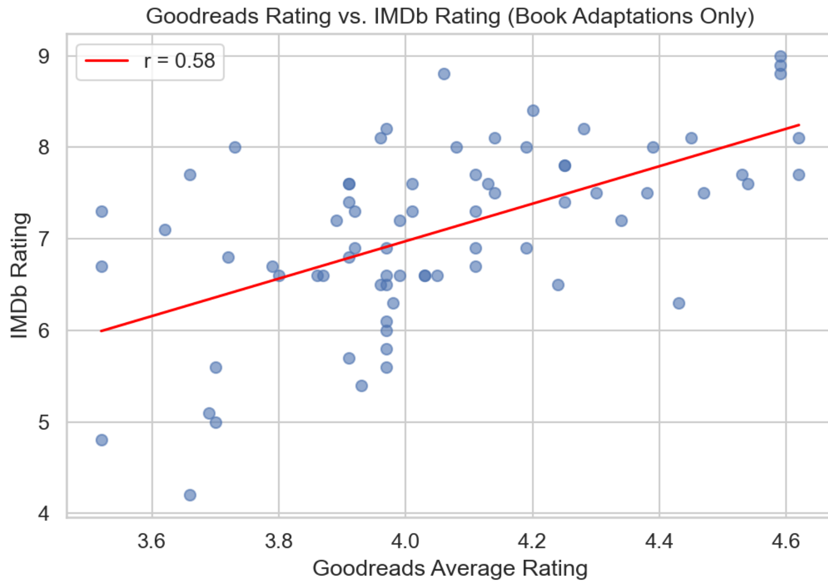

Figure 11: Goodreads vs IMDb

4 Statistical Models

To move beyond pairwise correlations, we fit two ordinary least squares (OLS) regression models, one predicting IMDb rating and one predicting log(domestic gross), using the same set of film-level predictors. Genre was coded with Action as the reference category (the most common genre). MPAA rating used PG-13 as the reference (the most common rating). Release timing was captured via binary indicators for summer (May–July) and holiday (November–December) release windows. Runtime was included as a continuous predictor. Decade of release was included as a continuous variable to account for temporal trends.

After listwise deletion of rows missing any predictor, the model sample consisted of **487 films**.

4.1 Model 1: IMDb Rating

Model: `IMDb Rating ~ Runtime + MPAA + Genre + Summer + Holiday + Decade`

Predictor	Coefficient	SE	t	p
Intercept	23.02	5.95	3.87	< 0.001
Runtime (per min)	**0.011**	0.002	6.85	< 0.001
MPAA: G (vs PG-13)	**+0.663**	0.182	3.64	< 0.001
MPAA: PG (vs PG-13)	**+0.229**	0.097	2.36	0.019
MPAA: R (vs PG-13)	**+0.251**	0.105	2.39	0.017
Genre: Biography (vs Action)	**+0.534**	0.209	2.55	0.011
Genre: Comedy (vs Action)	−0.211	0.119	−1.77	0.078
Genre: Adventure (vs Action)	+0.059	0.103	0.58	0.563
Genre: Drama (vs Action)	+0.079	0.144	0.55	0.583
Summer release	−0.062	0.079	−0.79	0.429
Holiday release	+0.003	0.095	0.04	0.972
Decade	−0.009	0.003	−2.94	0.003

$R^2 = 0.184$, Adjusted $R^2 = 0.163$, F(12, 474) = 8.90, p < 0.001, n = 487

Interpretation: Runtime is the most statistically significant predictor of IMDb rating. Each additional minute of runtime is associated with a **+0.011 point** increase in rating (p < 0.001). Controlling for other factors, G-rated films rate **+0.66 points** higher than PG-13 films, and R-rated films rate slightly higher (+0.25). This may reflect that R-rated films in the top 500 tend to be more critically focused. Biography films rate **+0.53 points** higher than Action films. Decade has a small negative effect, meaning more recently released films tend to have slightly lower IMDb ratings.

Release timing (summer or holiday) shows no significant effect on IMDb rating after controlling for other variables.

The overall model explains **18.4% of variance** in IMDb rating (adjusted $R^2 = 0.163$). The compressed rating range in this sample limits how much variation can be explained.

4.2 Model 2: log(Domestic Gross)

Model: `log(Domestic Gross) ~ Runtime + MPAA + Genre + Summer + Holiday + Decade`

Predictor	Coefficient	SE	t	p
Intercept	7.62	2.97	2.57	0.010
Runtime (per min)	**+0.0047**	0.001	5.79	< 0.001
MPAA: G (vs PG-13)	+0.085	0.091	0.94	0.348
MPAA: PG (vs PG-13)	+0.057	0.048	1.19	0.235
MPAA: R (vs PG-13)	−0.127	0.052	−2.44	0.015
Genre: Biography (vs Action)	−0.316	0.104	−3.02	0.003
Genre: Comedy (vs Action)	−0.176	0.059	−2.96	0.003
Genre: Drama (vs Action)	−0.128	0.072	−1.78	0.076
Genre: Adventure (vs Action)	+0.035	0.051	0.67	0.501
Summer release	+0.064	0.039	1.64	0.101
Holiday release	+0.067	0.048	1.41	0.159
Decade	**+0.0055**	0.001	3.72	< 0.001

$R^2 = 0.186$, Adjusted $R^2 = 0.166$, F(12, 474) = 9.05, p < 0.001, n = 487

Interpretation: Runtime is again a strong predictor. Each additional minute is associated with a **0.47% increase in domestic gross** (since $e^{0.0047} \approx 1.005$). R-rated films earn significantly less than PG-13 films, about **12.7% lower** on the log scale. Biography films earn about **27% less** than Action films, and Comedy films earn about **16% less**. Decade has a positive effect, meaning more recent films earn more in nominal terms.

Release timing shows a positive but not statistically significant effect after controlling for other variables.

The model explains **18.6% of variance** in log(gross), similar to the IMDb model. This suggests that other factors not included in the model likely play a large role in box office success.

4.3 Model 3: Elastic Net

The elastic net results align closely with the ordinary least squares findings. Runtime, genre, and vote count consistently emerge as the most influential predictors for both outcomes, demonstrating stable explanatory value. In contrast, weaker predictors are penalized and shrunk to zero. The adaptation indicator (`is_book`) is excluded from the final model, indicating that, after accounting for other film characteristics, adaptation from a book does not provide meaningful predictive power.

Model performance is evaluated using predicted versus actual plots for both IMDb rating and log-transformed domestic gross. In both instances, predictions follow the general upward trend of the data, although there is considerable dispersion around the identity line. This pattern suggests that the model captures broad trends but leaves a significant portion of variation unexplained. Cross-validation results further indicate that the chosen level of regularization minimizes prediction error and prevents overfitting.

In summary, the elastic net model supports the conclusions established by the OLS analysis. The consistency of key predictors across modeling approaches indicates that the findings are robust and not dependent on model specification. However, the moderate predictive accuracy underscores the complexity of modeling film success and the impact of unobserved factors not included in the dataset.

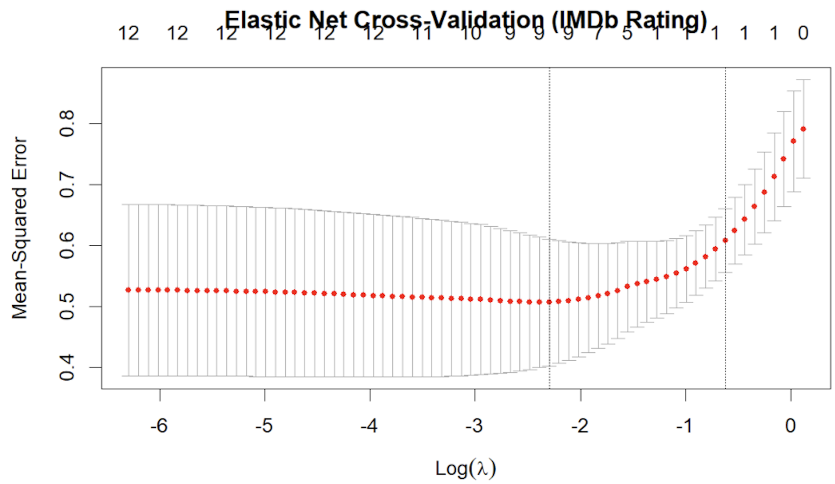

Figure 12: Cross Validation

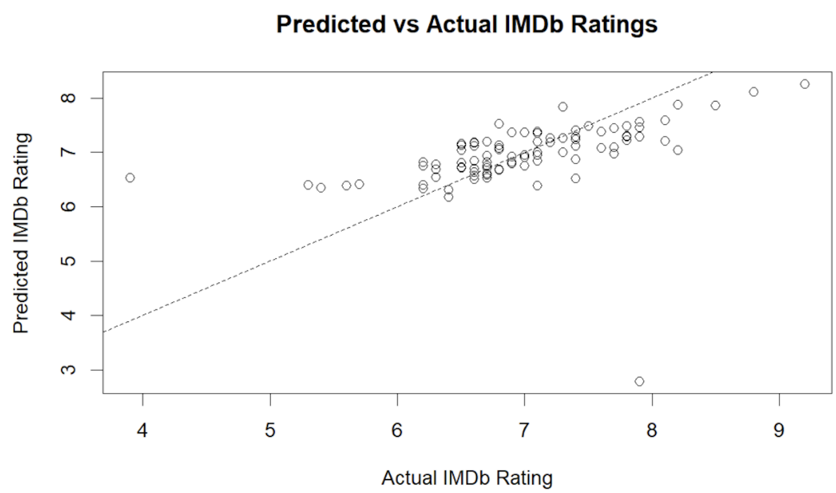

Figure 13: Predicted vs Actual IMDB Ratings

5 Conclusion

This analysis examined how film characteristics relate to both critical success (IMDb rating) and commercial success (domestic box office gross) within the top 500 highest-grossing films.

There are several limitations present throughout the analysis. Our population has been limited to the top 500 highest-grossing films due to data constrictions so our analysis can only be applied to that population. Our results reflect variation within high-performing movies.

There were several omitted variables such as budget and market intensity, franchise and branch effects, release strategy and competition.

We had a measurement constraint involving our adaptation being modeled as a binary indicator. This doesn't capture source material quality or audience familiarity.

Moving onto our results, the patterns present are consistent across all outcomes. Runtime is the strongest predictor in the models, with longer films tending to receive higher ratings and earn more revenue. This likely reflects the types of films that are given longer runtimes, such as large-scale or high-budget productions, rather than runtime itself directly causing success.

Genre plays an important role, but its effects differ across outcomes. Biography films tend to receive higher IMDb ratings but earn less at the box office, while Action and Adventure films dominate commercially. This suggests that films that appeal to broader audiences are more successful financially, even if they are not rated as highly.

MPAA rating also shows different effects. PG-13 films perform best commercially, likely due to wider audience access. In contrast, R-rated films receive slightly higher ratings but earn less, indicating a tradeoff between audience size and perceived quality.

Release timing does not have a significant effect after controlling for other variables. Although many top films are released during summer or holiday periods, this appears to be explained by other factors such as genre and budget.

Book adaptations do not show a clear advantage in either ratings or revenue. However, they are longer on average, and within this group, higher-rated source material is associated with higher IMDb ratings. This suggests that the quality of the underlying book may matter for audience reception, even if adaptation status itself does not.

Both models explain about 18% of the variation in outcomes. This indicates that while the included variables capture some important trends, a large portion of variation in film success is driven by factors not included in the model, such as marketing, franchise effects, or star power.

Overall, the results show that film success is influenced by a combination of structural characteristics and broader industry factors, and no single variable fully explains either ratings or revenue.